# AWANTIKA SRIVASTAVA


**AI Engineer | Machine Learning, Data Science & NLP**

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## PROFILE SUMMARY

AI Engineer with **2+ years of experience building Generative AI, LLM applications, and machine learning systems**. Skilled in developing **RAG pipelines, AI agents, and NLP solutions using Python, LangChain, and transformer models**. Experienced in building **end-to-end AI workflows, integrating vector databases, and deploying scalable AI microservices using FastAPI and cloud platforms**. Strong background in **machine learning pipelines, conversational AI, and production deployment of AI systems**.

## CORE TECHNICAL SKILLS

- **Programming Languages & Tools:** Python, C++, SQL.
- **Statistics & Mathematics:** EDA, Statistical Modeling, Hypothesis Testing, A/B Testing, Probability, Linear Optimization, Trend Analysis.
- **Machine Learning:** Supervised & Unsupervised Learning, Regression, Classification, Clustering, Random Forest, Decision Trees, SVM, KNN, K-Means, XGBoost, Hyperparameter Tuning.
- **Deep Learning:** Neural Networks, CNN, RNN, LSTM, Transformers (BERT), Transfer Learning, Model Fine-tuning, TensorFlow, Keras, PyTorch.
- **Generative AI & LLM:** LLM Applications, RAG Pipelines, Prompt Engineering, LangChain, CrewAI, AI Agents, Semantic Search, FAISS, Pinecone, Weaviate
- **Conversational AI:** STT (Speech-to-Text), TTS (Text-to-Speech), Voice AI Integration, Conversational AI Systems
- **Computer Vision**: Image Classification & Preprocessing, Object detection (YOLO, SSD, MobileNet, ResNet), Video Analytics, OpenCV
- **NLP**: Text preprocessing, Tokenization, Chunking, Sentiment Analysis, Transformer-based models, NLTK, TF-IDF, SpaCy
- **Backend & APIs:** FastAPI, Flask, REST APIs, Microservices, n8n
- **MLOps & Deployment:** Docker, Git, CI/CD, Model Monitoring, AWS (EC2, SageMaker), GCP (Exposure), Linux

## EXPERIENCE

**AI/ML Engineer | PPS International Pvt. Ltd.** **January 2024-Present**

- Built **LLM-powered applications and RAG pipelines** for document search and question answering systems.
- Designed **AI agents and multi-agent workflows** for automating information retrieval and decision support tasks.
- Integrated **vector databases (FAISS / Pinecone)** to enable semantic search and efficient retrieval of knowledge base documents.
- Developed **FastAPI-based microservices** to serve AI models and LLM applications in production environments.
- Implemented **prompt engineering and evaluation frameworks** to improve the accuracy and reliability of LLM responses.
- Designed **end-to-end AI pipelines including data ingestion, embedding generation, vector indexing, and LLM inference**.
- Collaborated with cross-functional teams to integrate **AI workflows and automation pipelines** into enterprise systems.
- Deployed AI solutions on **cloud environments using AWS services** for scalable production use cases.

## PROJECTS

**Railway Driver Assistance System (RDAS) | Enterprise ML Project**

- Designed and deployed a **real-time ML systems** for unsafe behavior detection using **CNN-based object detection** models.
- Implemented **end-to-end ML pipelines** from data ingestion to production inference.
- **Deployed** optimized models using **TensorFlow Lite** for continuous, **low-latency inference.**
- Developed a **Flask-based** web interface to display detections and recorded video clips in **real time**.
- **Deployed** the complete solution on **edge devices**, enabling reliable on-device inference and **real-time event recordings**.

**Amazon Stock Price prediction | Applied ML Project**

- **Built batch-oriented ML workflows** for time-series forecasting using **LSTM** models.
- Implemented data **preprocessing**, feature engineering, and model **evaluation pipelines** on **large** historical datasets.
- Designed **sliding-window** based **sequence generation** and trained LSTM model 5-year stock price data.
- Evaluated model performance using appropriate regression **metrics** and **trend-based analysis** for short-term forecasting.

**YouTube Comments Sentiment Analyzer | link - https://youtube-ai-analyzer-ndzqo6r2mepjrtsdjmwaxl.streamlit.app/**

- Deployed transformer-based **NLP models** as **production-ready** services with **REST APIs**.
- **Fine-tuned** and served a **DistilBERT**-based **sentiment** classification model for large-scale text inference.
- **Built** and deployed an interactive **streamlit web application** to perform real-time **sentiment analysis** on YouTube comments.
- **Processed** high-volume text **data** with sub-second inference **latency** for real-time sentiment analysis.

## CERTIFICATION

- IBM Data Science & AI Certification
- AWS Generative AI with Large Language Models
- OpenCV Computer Vision Certification

## EDUCATION

**IMS Engineering College, Ghaziabad** **September - 2020**

**Bachelor of Technology (Electrical and electronics engineering)**